Foldback handoff

PayPal AI Hackathon entry. Written 29 October 2026. Submission deadline: Thursday 12 November 2026, 12:00 PT (target: submit 11 November). Judging 1 to 15 December 2026.

1. What Foldback is

A group payment where each friend's share is a PayPal authorization hold. If the headcount is met by the deadline, every hold is captured. If not, every hold is voided: **nobody pays unless the group makes it**. An LLM reads a pasted group chat into a structured plan, an organizer assistant simulates, drafts and proposes, and **code (domain/decide) makes every money decision**. The organizer sees a Bryntum Scheduler board that updates live over Socket.IO. Goals, equal weight: win the hackathon, and ship a production-grade system.

2. Plan change (decided 29 October)

The backend, Postgres and the public tunnel or proxy run on a friend's Mac mini. The Next.js frontend is deployed on Vercel. The three-account Azure plan (database on Nishit's account, VM on a teammate's account, spare VM) is dropped. Reason: the Mac mini has far more compute than a B1ms or B2s Azure machine, it removes the credit and region limits, and Vercel gives the frontend a global CDN and automatic HTTPS for free.

Topic	Before (Azure)	Now (Mac mini)
Database	Azure Database for PostgreSQL Flexible Server, separate from the app	Postgres 16 in Docker on the Mac mini, data on a named volume
Backend host	Ubuntu VM with Docker Compose	Docker Compose on macOS (Docker Desktop or OrbStack): Postgres, backend, tunnel or proxy
Frontend host	A container behind Caddy on the same VM	Vercel, built from GitHub. No frontend Dockerfile or Compose service needed
CPU architecture	linux/amd64 only	linux/arm64 if the Mac is Apple Silicon. CI must build arm64 (or multi-arch)
Public access	Static IP and cloudapp.azure.com name, Caddy and Let's Encrypt	A tunnel (Tailscale Funnel or Cloudflare Tunnel) or router port-forward plus Caddy. To decide, see section 7
Deploy	CI pushes over SSH to the VM	CI builds images to ghcr.io; the Mac pulls them (a self-hosted runner or a pull script). To decide
Backups	Azure automatic 7 days plus pg_dump	pg_dump daily to a second location (external disk plus an off-site copy). The Mac is now the only copy otherwise
Scale-out	Second VM on another account	Dropped. One machine is enough for a demo

Specs still saying Azure that need rewriting: `docs/spec/deploy.md`, `phases/03-backend-foundation.md` step 7 (sub-phase 3f), the Azure notes in `architecture.md`, and `docs/team-setup.md`. The database code does not change: it already uses two URLs (app role and owner role) and a direct, non-pooled connection.

3. Status

Phase	State	Notes
0 Spike	Done	PayPal authorize, capture and void proven in the sandbox. Free LLM chain tested. Bryntum board renders under webpack
1 Database design	Done	Schema, migrator, 31 schema tests, reviewed and merged
2 UI design	Done, merged	Clickable HTML mockups in docs/design/. Tokens, logo, five screens, light and dark theme, scroll scenes
3a Backend foundation	Built, not committed	Branch phase3a-backend-foundation. 52 tests, typecheck, lint and format pass

Phase	State	Notes
3b to 3e	Next	Transactions and locks, auth and seeds, contracts read and Socket.IO, quality tooling and Dockerfile and CI
3f Deploy	Needs the plan change	Rewrite for the Mac mini before starting
4 to 9	Later	Rule reader (LLM bake-off), frontend and board, pay and hold, resolve, agent, ship (README, video, Devpost)

4. Decisions already made

- **Stack:** Node 24, Yarn 4.9.2 via Corepack (nodeLinker node-modules, enableScripts false), TypeScript strict, Express 5, Postgres 16 with plain pg and numbered SQL migrations (checksummed), zod 4, Socket.IO (invalidation-only events), Next.js 16 App Router with `next dev/build --webpack` (Turbopack crashes on Bryntum), MUI 7, Redux Toolkit and RTK Query, Jest 30.
- **Money:** PayPal Orders v2 with intent AUTHORIZE, then capture (final_capture true) or void. Idempotency keys per operation. Unknown results are stored and synced, never guessed. Sandbox only: PAYPAL_ENV must be exactly sandbox or the server refuses to boot.
- **LLMs, free tiers only:** a fallback chain (Gemini Flash models, then Groq gpt-oss-120b and 20b, then OpenRouter free models) called with plain fetch. No paid provider, no card-backed key, no SDK. Daily and per-minute call caps. The code rejects a non-free OpenRouter model at boot.
- **Database rules:** constraints and triggers live in the schema. The app role has DML only. Never delete live money. Events are append-only. Session advisory locks need a direct (non-pooled) connection.
- **Design:** sharp corners (radius 0), no serif, IBM Plex Sans and Plex Mono, sky blue #60CDFF with black, blue #0070E0 and navy #012169. Status is always a word. Light theme by default with a footer toggle, applied on every page. Logo: Fold (black on sky, fold top-right) and back (white on black, fold bottom-right).
- **Frontend standard:** one Redux Toolkit store, one baseApi extended per feature, one status map, shared primitives. Folder layout: `components/ComponentName/index.tsx` plus `styled.ts` plus `index.test.tsx`.

5. Repository map

Path	What is there
docs/spec/	The specification. Start at README.md, then quality.md (permanent rules) and the active phases/NN file
docs/design/	HTML mockups and design tokens (tokens.css is the source of truth)
backend/	Express API, migrations, tests. src/ follows routes, controllers, services, repositories, db
deploy/	compose.dev.yml for local Postgres. Production Compose, Caddyfile and scripts are created in 3e and 3f
.claude/tasks/, learn/	Local only (gitignored): the per-phase plans and the numbered study notes

6. Working rules

- Never put AI attribution in commits, pull requests, code comments or docs. Nishit pushes and merges; branches are deleted after the merge.
- Never commit a secret. A secret scanner runs on pull requests (it once flagged made-up PayPal-style ids in mock data, so keep sample ids obviously fake).
- Before coding: read the phase file, write a plan, search before adding anything, and use only approved dependencies (docs/spec/quality.md section 3). Gates after each change: format:check, typecheck, lint, tests (knip, jscpd and the rest arrive in 3e).
- Money, auth or deletion changes get a second adversarial review before merge.

- Git is done one command at a time: the user pastes the output and gets the next command.

Run the backend tests locally:

```
docker compose -f deploy/compose.dev.yml up -d
cd backend; corepack yarn install
corepack yarn test
```

TEST_DATABASE_URL defaults to port 5433 (the throwaway test database in the Compose file). Copy backend/.env.example to .env for the server; JWT_SECRET must be at least 32 characters.

7. Frontend on Vercel, backend on the Mac mini: one design decision

The browser will load the app from a Vercel address and call an API on the Mac mini's address. These are different sites, so the session cookie (fb_session, SameSite=Lax) would not be sent to the API, the Next.js middleware could not see it, and the Socket.IO handshake would fail. Two ways to fix it:

Option	How it works	Cost and risk
A. Same-origin through Vercel rewrites (recommended to start)	next.config rewrites /api/* to the Mac's public HTTPS address. The browser only talks to the Vercel address, so cookies and the Origin check behave as designed. Socket.IO connects straight to the Mac with a short-lived token from a new endpoint, because Vercel does not proxy WebSockets	Free, no domain. One new endpoint (socket token) and a small change to the Socket.IO handshake. PayPal return URLs and CORS_ORIGINS use the Vercel address
B. One parent domain	Buy a domain, for example app.example.com on Vercel and api.example.com through a Cloudflare Tunnel, and set the cookie Domain to the parent	About 10 USD a year. Simplest runtime behavior, needs the domain on Cloudflare

Either way: the Vercel environment variables hold only the public API address, never a secret. Bryntum's trial package installs from public npm, so the Vercel build needs no extra credentials. Vercel's Hobby tier is for non-commercial use, which fits a hackathon.

8. Mac mini hosting: open questions for the friend

Question	Why it matters
Which Mac mini (chip, RAM, free disk) and which macOS version?	Apple Silicon means linux/arm64 images. Postgres and Docker need a few GB of RAM and disk
Will it stay on and awake from now to 15 December? Is it on a wired connection? Is there a UPS?	Sleep, updates and power cuts take the demo offline. Disable sleep, schedule updates, and turn on start-after-power-failure
Can the friend run a tunnel (Tailscale Funnel or Cloudflare Tunnel)?	Judges and the PayPal return URL need a public HTTPS address. Tailscale Funnel gives a free stable https name without a domain. Cloudflare Tunnel needs a domain for a stable name. Router port-forwarding often fails behind carrier-grade NAT
Docker: Docker Desktop or OrbStack? Starts at login?	Compose must come back after a reboot (restart: unless-stopped plus auto-start at login)
Who has SSH or remote access, and how do deploys reach the machine?	CI needs a way to deploy. A self-hosted GitHub runner on the Mac, or a pull script triggered by a webhook, avoids opening SSH to the internet
Where do backups go besides the same disk?	A single machine is a single point of loss. Daily pg_dump to an external disk plus an off-site copy

Risk to be honest about: a home machine is less reliable than a cloud VM. Keep the demo video as the judging artifact, record it before the deadline, and keep the Neon free database or a cloud fallback in mind if the Mac is unreachable near 12 November.

9. Next steps

- Commit and push 3a (branch phase3a-backend-foundation), merge, clean up the branch.

- Rewrite deploy.md and phase 3f: Mac mini for Postgres, backend and the tunnel (arm64 builds, backups, deploy method), Vercel for the frontend, and the cross-site decision in section 7. Update team-setup.md and phase 5 (no frontend Dockerfile).
- Continue 3b (transaction and lock helpers), 3c (auth, seeds, reset), 3d (contracts read, Socket.IO), 3e (quality tooling, Dockerfile, CI).
- Then 3f deploy on the Mac mini, and Phase 4 (rule reader) once a free LLM key is set per person.
- Add the teammates as GitHub collaborators and send them docs/team-setup.md (still pending).